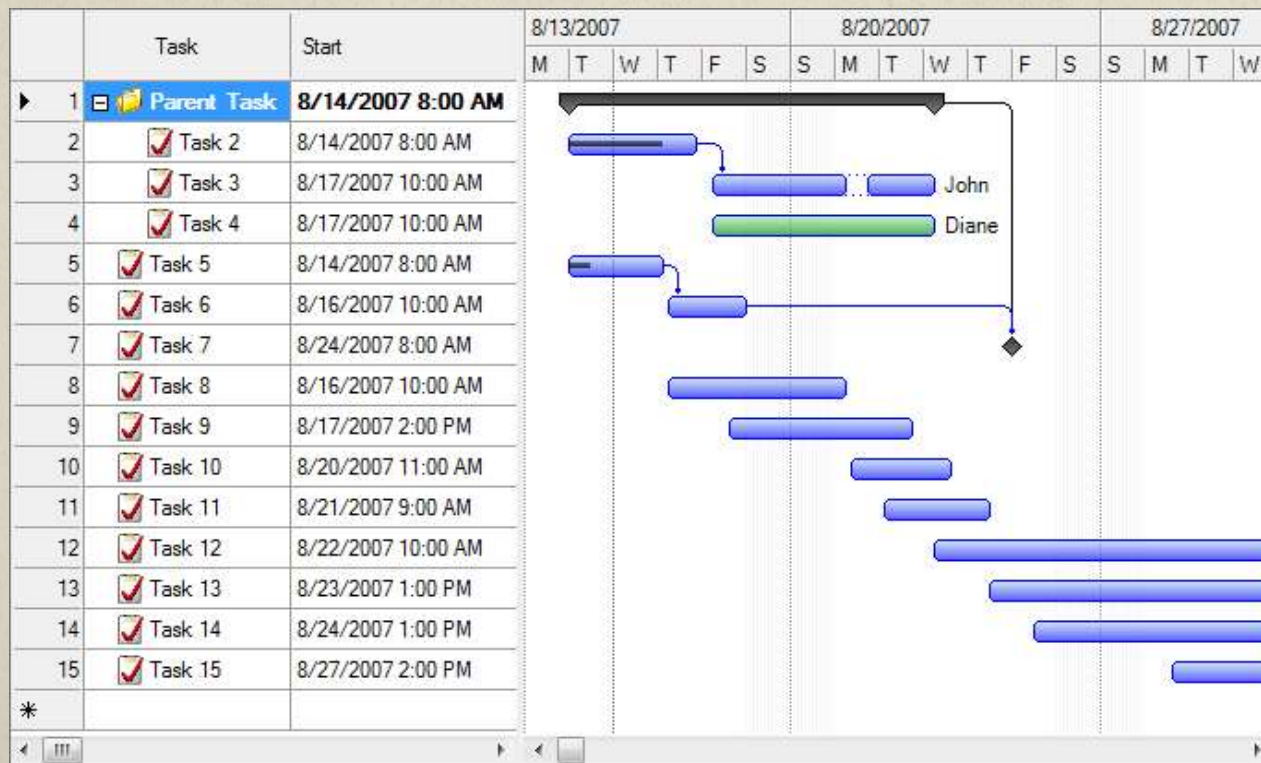


PROJECT SCHEDULING

MGMT-3105





Online PM Courses
Build Your Project Career

WHAT IS A GANTT CHART?

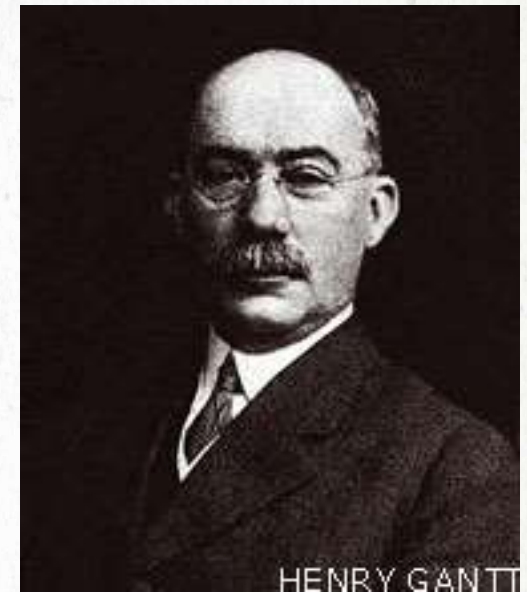


GANTT CHARTS

- Developed by Henry Gantt (1861-1919) between 1910 and 1915, they are another extremely useful tool for showing a project network.
- Gantt charts establish a time-phased network, which links project activities to a project schedule
- You need to understand the appropriate use of them and how to read/make them.

On which project was the Gantt chart heavily utilized for the first time between 1931 and 1936?

The Hoover Dam *(Nevada–Arizona border)*



GANTT CHARTS

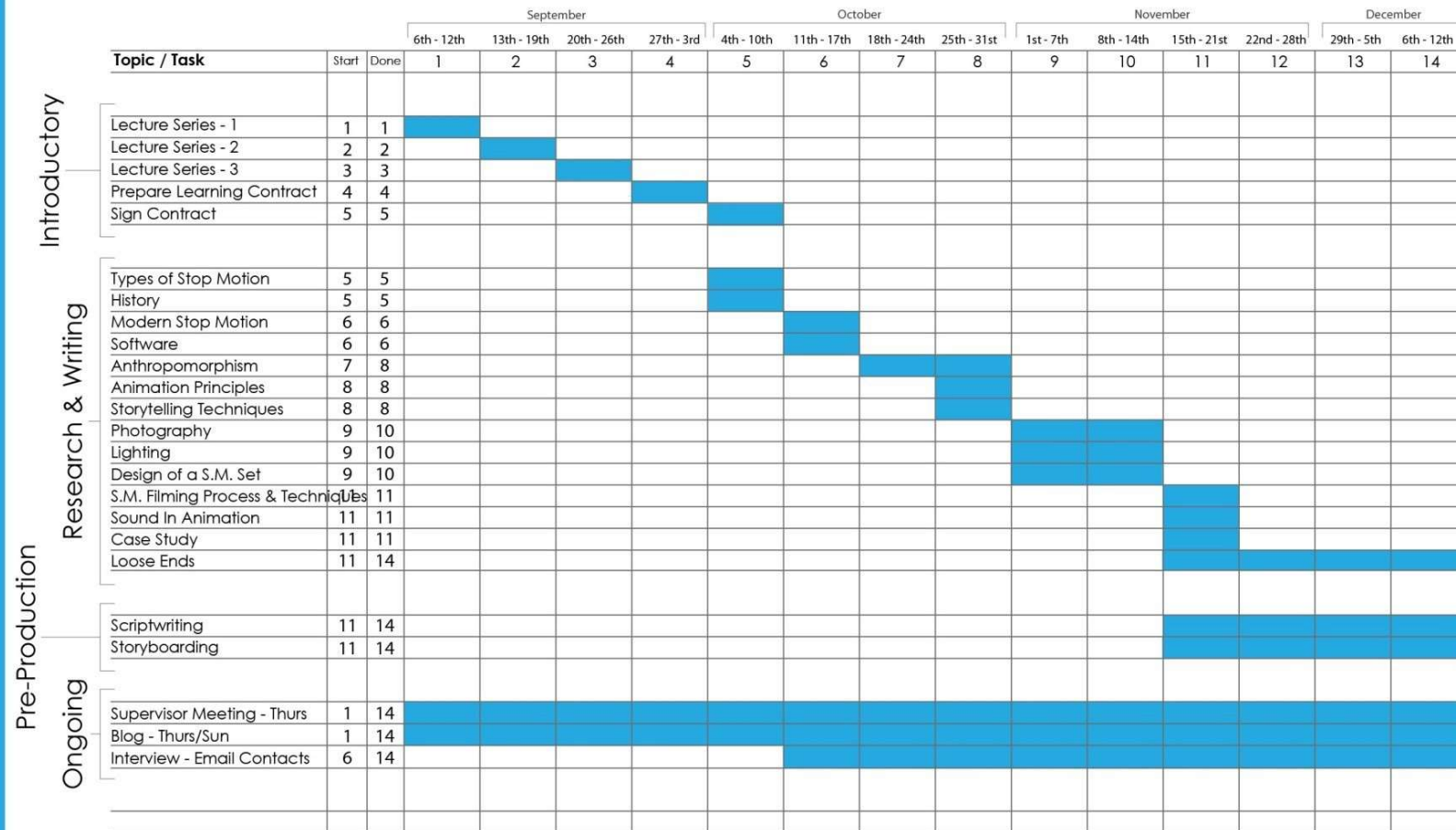
When they are linked to a project schedule, they can be used to track the progress of the project against its plan.

Some of the benefits of Gantt Charts are:

- a) They are easy to read and comprehend
- b) They identify the project network coupled with the schedule
- c) They allow for updating and project control
- d) They are useful for assigning resources to tasks
- e) They are also easy to create.

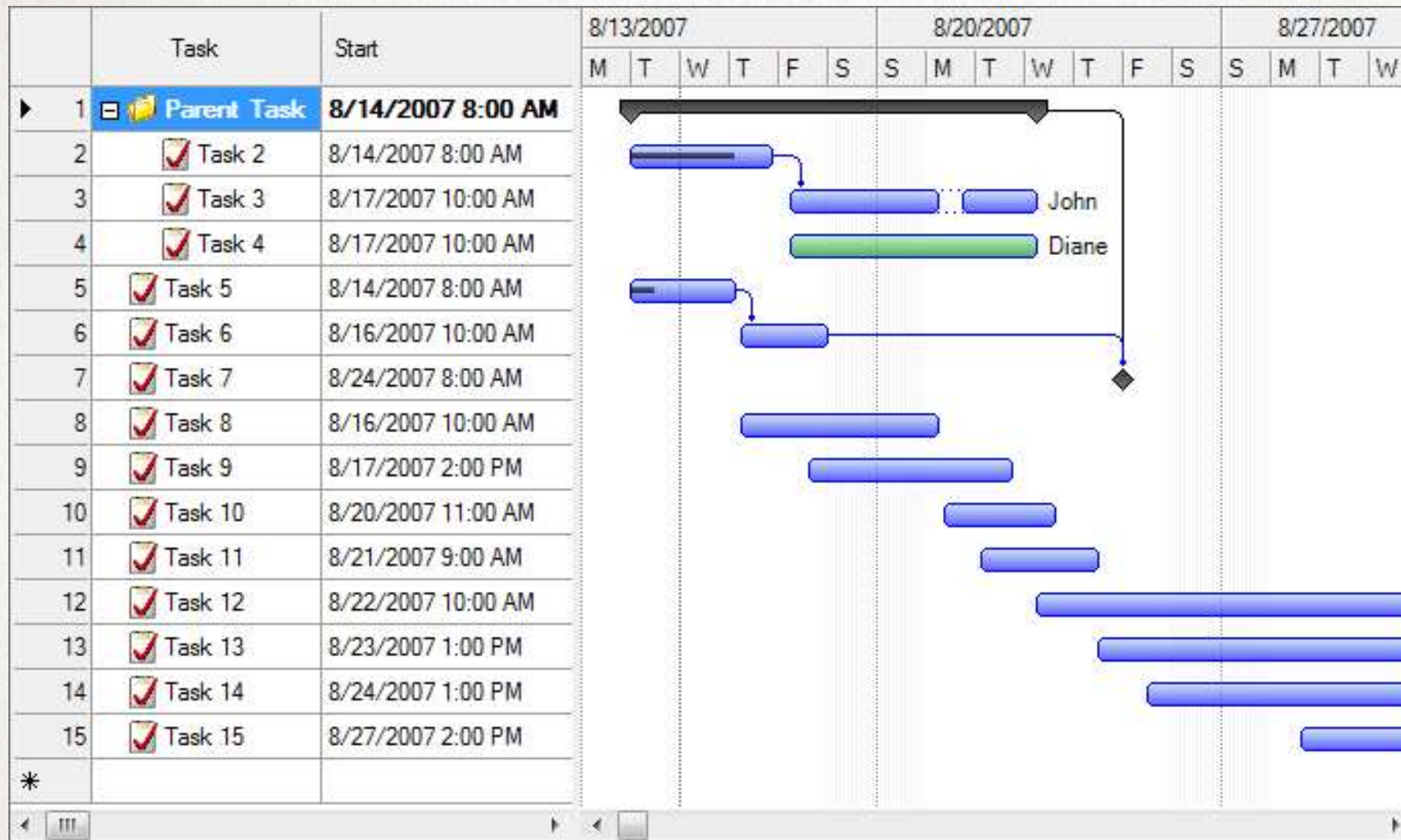
EXAMPLE GANTT CHART FORMATS

Gantt Chart HONOURS PROJECT 2010/11



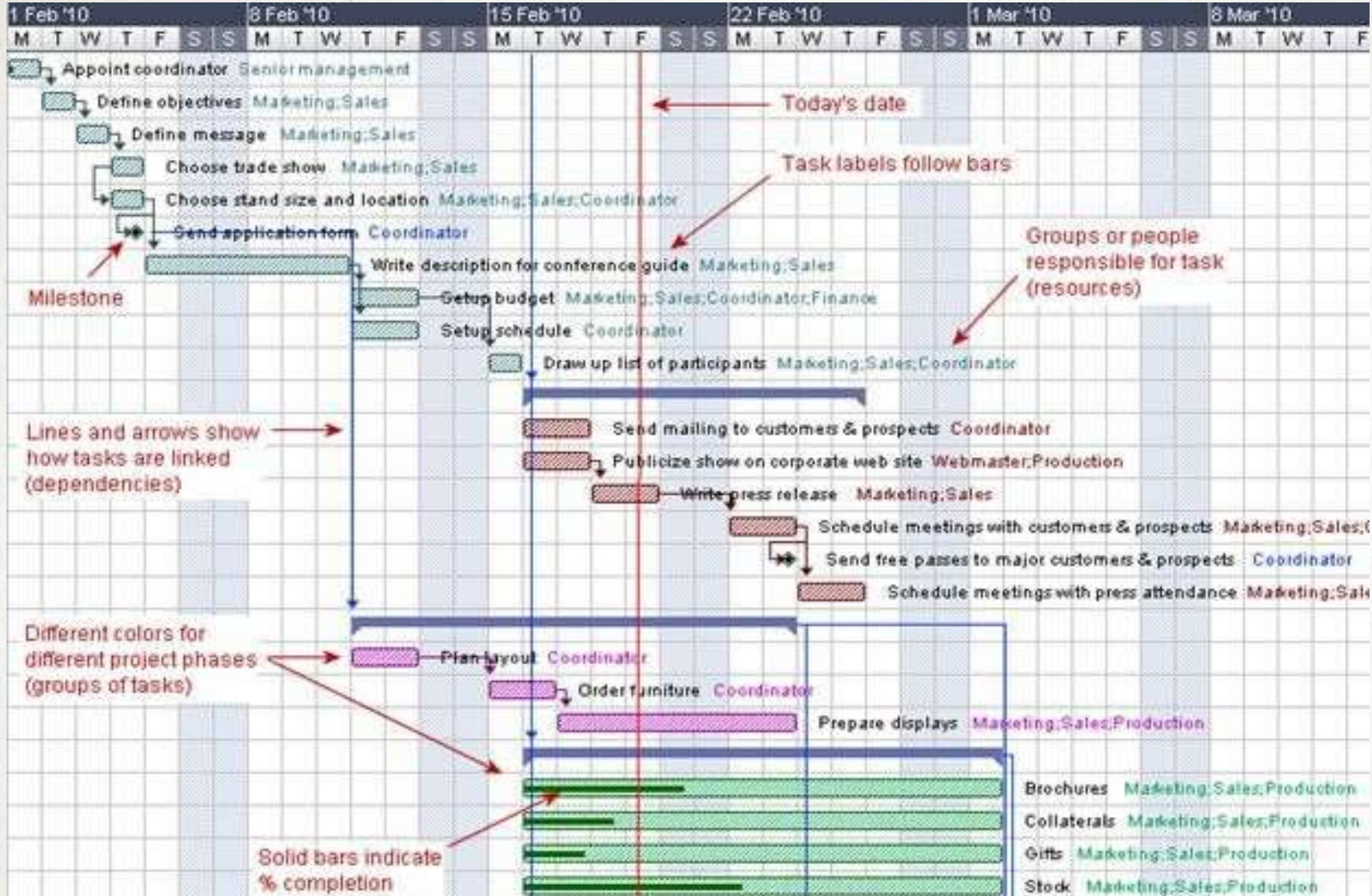
Simple
No Resources

EXAMPLE GANTT CHART FORMATS



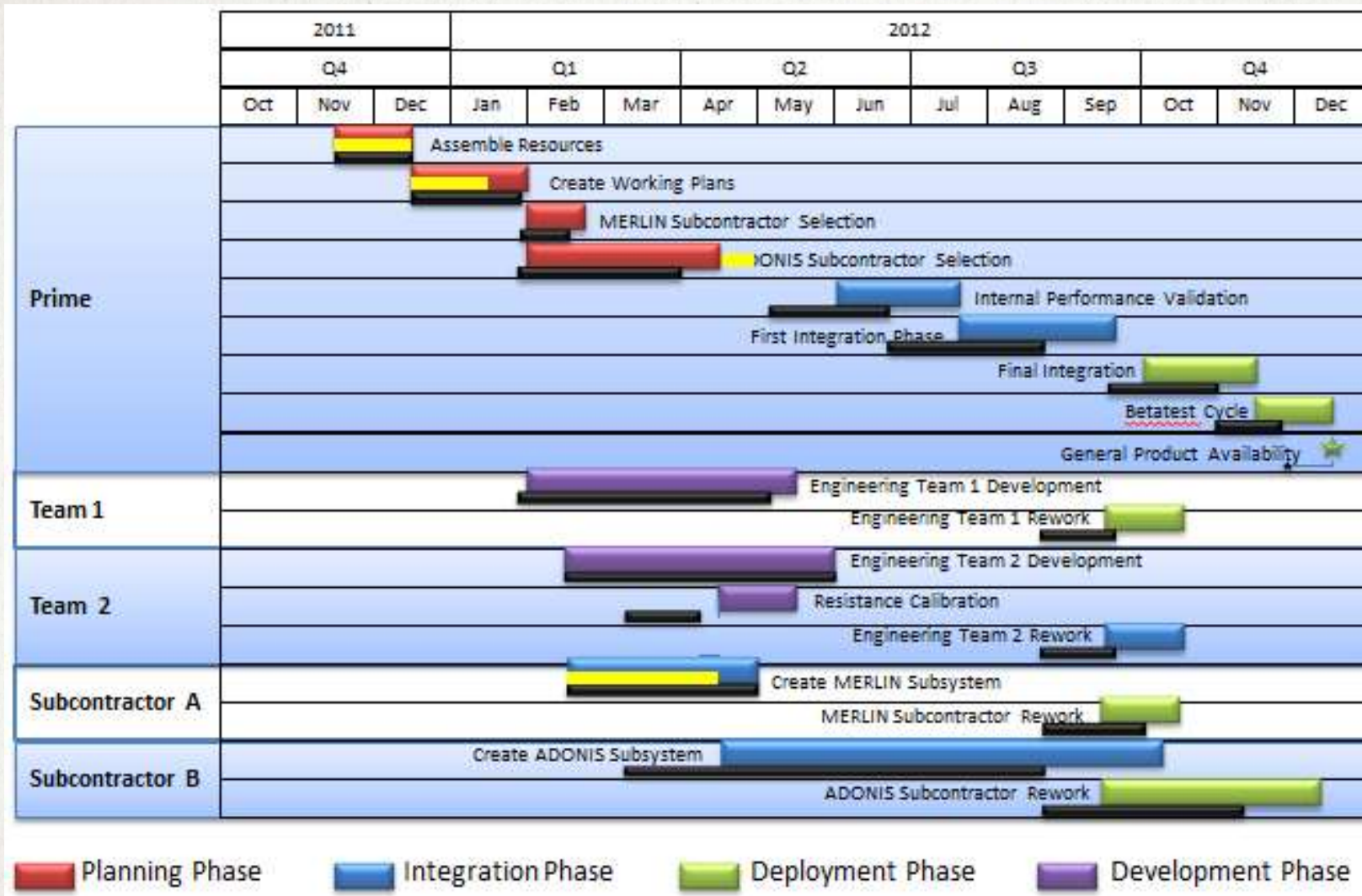
Now Add
Resources

EXAMPLE GANTT CHART FORMATS



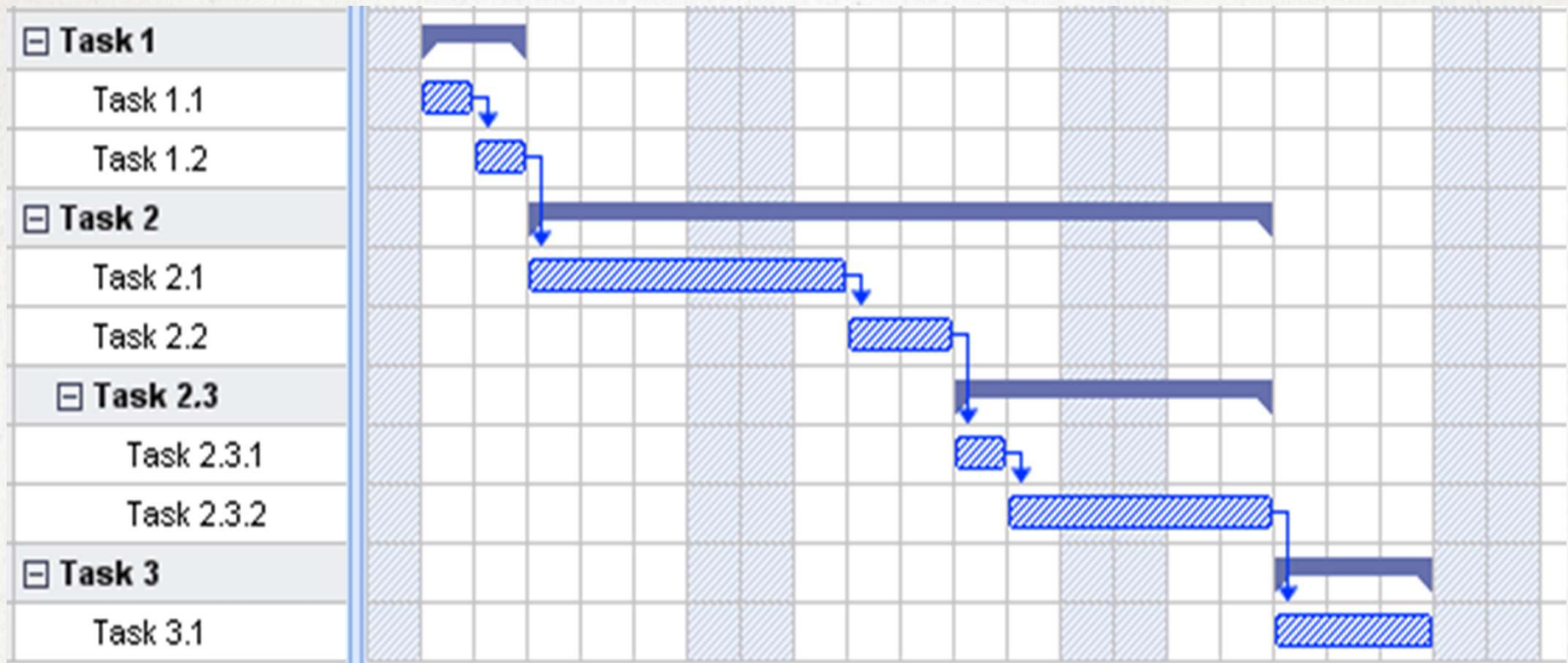
Now Get
Really
Crazy!

EXAMPLE GANTT CHART FORMATS



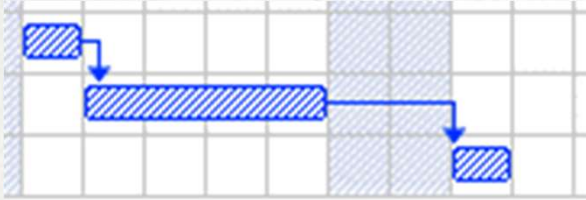
Completely Different.

GANTT CHART FORMATS

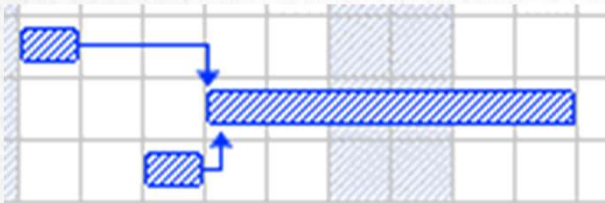


Gantt charts allow you to show precedence and successors within tasks.

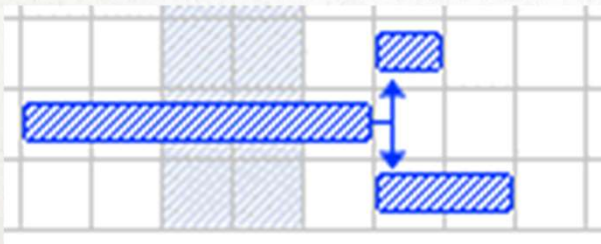
GANTT CHART FORMATS



Arrows show the order tasks must be completed



Tasks can have more than one preceding task.



Completing a task can lead to the Start of two or more tasks.

PRECEDENCE RELATIONSHIPS

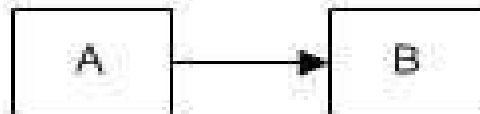
Finish To Start (FS) Relationship:

The second activity can not be started until the first activity completes. It is the most used dependency.

Example:

To paint a wall first you need to build a wall. In this case, first activity is building the wall and second activity will be painting. You can not start painting the wall unless the wall is ready

Finish To Start (FS)



PRECEDENCE RELATIONSHIPS

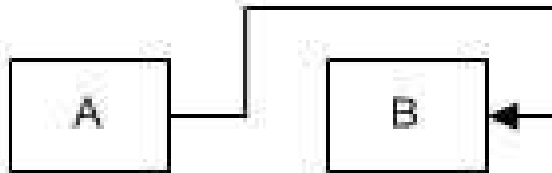
Finish To Finish (FF) Relationship:

The second activity cannot be finished until the first activity finishes.

Example:

You can not finish editing a paper until the paper has finished being written.

Finish To Finish (FF)



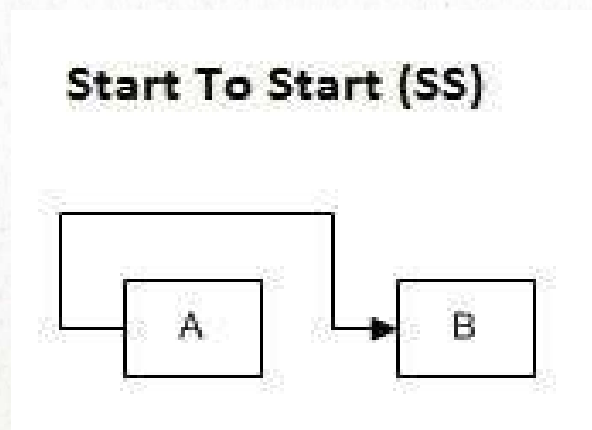
PRECEDENCE RELATIONSHIPS

Start To Start (SS) Relationship:

The second activity cannot be started until the first activity starts; both activities should start simultaneously.

Example:

To cut the grass you need to trim around areas the lawn mower can not get to as well as cut the lawn. Both activities can be started at the same time.



PRECEDENCE RELATIONSHIPS

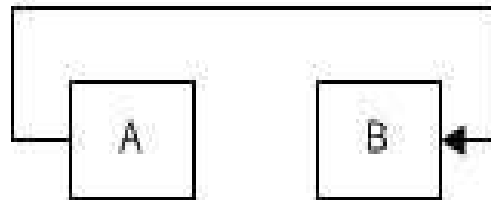
Start To Finish (SF) Relationship:

The start of the first activity triggers the finish of the second activity. **(VERY UNCOMMON)**

Example:

The start of the new shift signals the finish of the current shift.
Current shift can't end until the new one starts.

Start To Finish (SF)



CRASHING PROJECTS (*UNDER WHAT CONDITIONS*)?

- The initial schedule may be too optimistic.
- Market needs change and the project is in demand earlier than anticipated.
- The project has slipped considerably behind schedule.
- The contractual situation provides even more incentive to avoid schedule slippage.



OPTIONS FOR ACCELERATING ACTIVITIES

- Improving the productivity of existing project resources (*team meeting to focus efforts on the most important aspect of a task. Review best practice how to do the task*)
- Changing the working method employed for the activity, usually by altering the technology and types of resources employed (*Change from drawing review to CAD model review*)
- Increasing the quantity of project resources, including personnel, plant, and equipment (*OT, extra shifts, expedite freight*)...

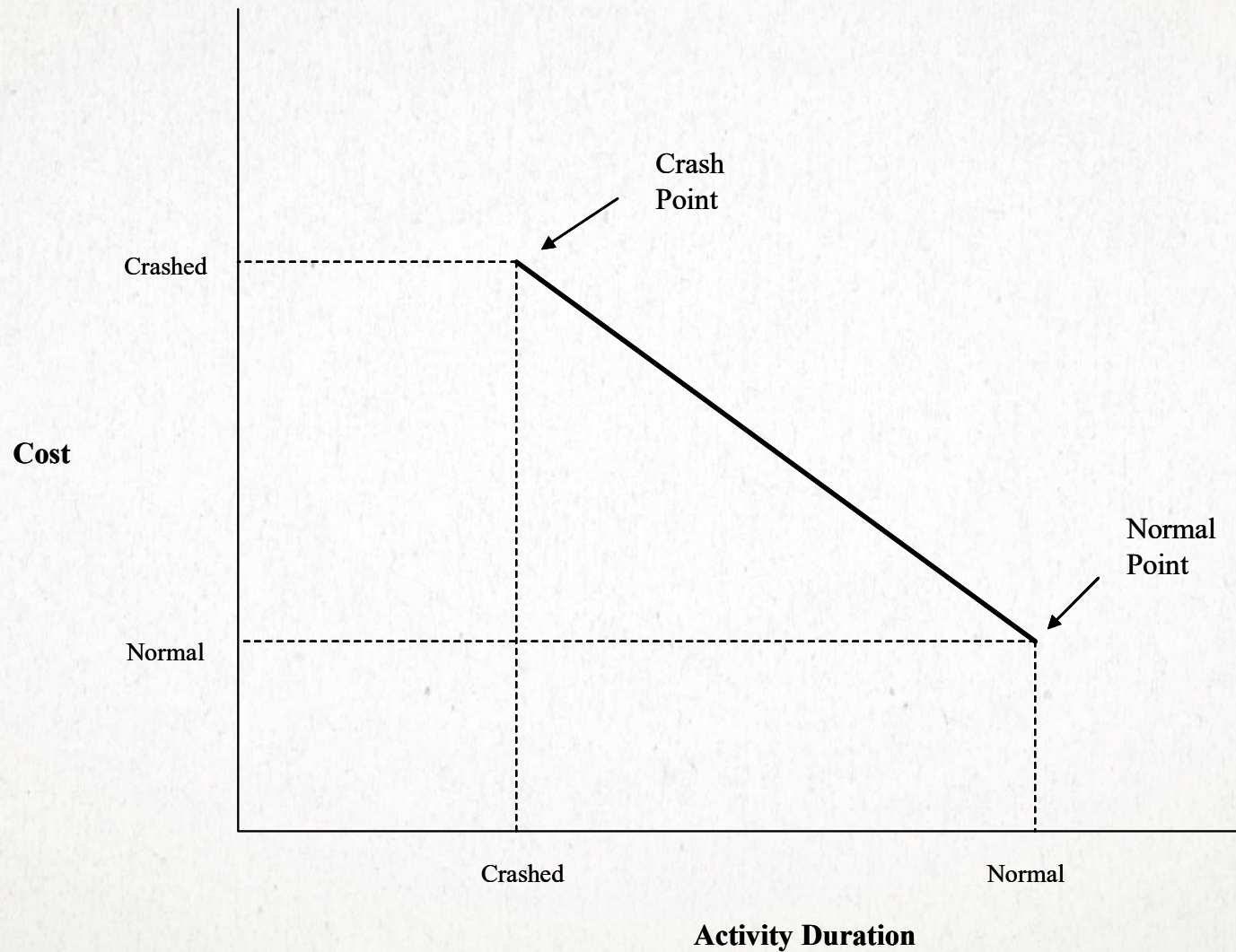
CAUTION WITH REQUESTING (*EXCESSIVE*) OVERTIME!

There is research that suggests that the greater the amount of overtime workers are expected to perform, the lower the marginal performance a company is likely to receive from them.

For example, it was found that in engineering functions, optimal performance was realized after only four hours of overtime per week.

By the time engineers were working **10 hours of overtime**, the marginal real extra output realized by the company had dropped to **zero!**

TIME-COST TRADE OFFS FOR CRASHING ACTIVITIES



CRASHING ACTIVITIES – AN EXAMPLE

Formula

$$Slope = \frac{\text{crash cost} - \text{normal cost}}{\text{normal time} - \text{crash time}}$$

SUPPOSE:

NORMAL ACTIVITY DURATION = 8 WEEKS

NORMAL COST = \$14,000

CRASHED ACTIVITY DURATION = 5 WEEKS

CRASHED COST = \$23,000

THE ACTIVITY COST SLOPE =

$$\frac{23,000 - 14,000}{8 - 5} \quad \text{or,} \quad \frac{\$9,000}{3} = \$3,000 \text{ per week}$$

CRASHING ACTIVITIES – AN EXAMPLE

Project cost changes when tasks along the critical path are being crashed.

The cost goes up for every task being crashed.

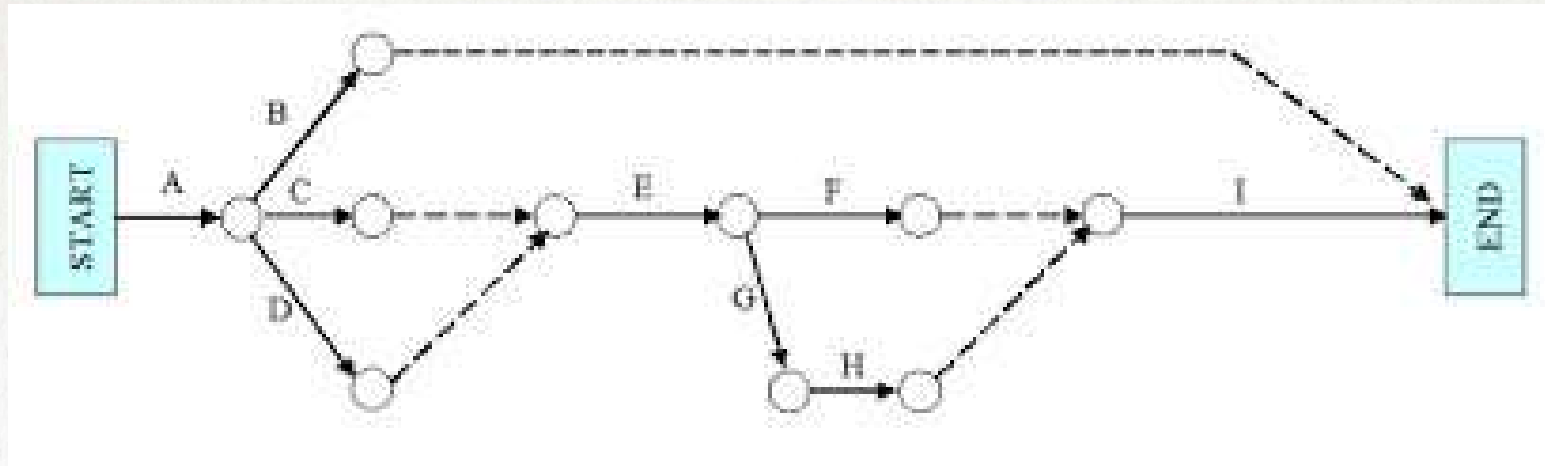
One can create a table listing all tasks, the normal duration and cost, and the crashed duration and cost.

Sample Project							
Activity	Normal				Crashed		
	Duration		Cost		Duration		Crashing Cost / day
A	3 days		\$ 1,000		2 days	\$ 1,500	\$ 500
B	4 days		\$ 700		2 days	\$ 1,000	\$ 150
C	3 days		\$ 2,500		2 days	\$ 4,000	\$ 1,500
D	6 days		\$ 1,500		6 days	\$ 1,500	N/A
E	9 days		\$ 3,750		7 days	\$ 9,000	\$ 2,625
F	4 days		\$ 5,000		3 days	\$ 7,000	\$ 2,000
G	6 days		\$ 2,550		5 days	\$ 3,500	\$ 950
H	8 days		\$ 5,500		4 days	\$ 7,550	\$ 513
Total:			\$ 22,500			\$ 35,050	

With this table it is possible to calculate the crashing cost per day for each task and select the task that makes the most financial sense to crash.

Tasks A, F and H are on the critical path.

THE NETWORK DIAGRAM



The above is an example of an Activity-on-Arrow diagram (AOA).

- Activity is on the arrow. Length of arrow denotes time. Easier to visualize passage of time.
- Dashed lines are dummy tasks – to show Finish-to- start.
- Node shows earliest and latest start.

DOWNSIDE TO NETWORK DIAGRAMS

1. May not be drawn to scale – time is just a number so can lead to a perception of more time than actually available.
2. Can be too large and complex to really be useful.
3. Faulty reasoning in network construction can sometimes lead to oversimplification or incorrect representations.
4. Networks are sometimes used for tasks which they are not well suited
5. When estimating there is a strong potential for positive bias in PERT estimations.



Create an AON diagram from this WBS

- 1 WBS for Software Implementation Project
 - 1.1 Project Management
 - 1.2 Product Requirements
 - 1.2.1 Software Requirements
 - 1.2.1.1 Draft Software Requirements
 - 1.2.1.2 Final Software Requirements
 - 1.2.1.3 Software Requirements Approval
 - 1.2.2 User Documentation
 - 1.2.2.1 Draft User Documentation
 - 1.2.2.2 Final User Documentation
 - 1.2.2.3 User Documentation Approval
 - 1.2.3 Training Program Materials
 - 1.2.3.1 Initial Training Requirements
 - 1.2.3.2 Initial Training Materials
 - 1.2.3.3 Trial Course Delivery
 - 1.2.4 Hardware
 - 1.2.4.1 Draft Hardware Requirements
 - 1.2.4.2 Final Hardware Requirements
 - 1.2.4.3 Hardware Requirements Approval
 - 1.2.5 Implementation & Future Support
 - 1.3 Detail Software Design
 - 1.3.1 Initial Software Design
 - 1.3.2 Final Software Design
 - 1.3.3 Software Design Approval
 - 1.4 System Construction
 - 1.4.1 Configured Software
 - 1.4.2 Customized User Documentation
 - 1.4.3 Customized Training Program Materials
 - 1.4.4 Installed Hardware
 - 1.4.5 Implementation & Future Support

<http://herdingcats.typepad.com/.a/6a00d8341ca4d953ef0133f339b8a9970b-pi>

First Draft AON Example

